

**FORM 2**

THE PATENTS ACT, 1970  
(39 of 1970)  
&  
THE PATENTS RULES, 2003

**COMPLETE SPECIFICATION**  
(See section 10 and rule 13)

**1. TITLE OF THE INVENTION**

**SYSTEM AND METHOD FOR MANAGING CHARGING  
INFRASTRUCTURE OF ELECTRIC VEHICLES THROUGH IOT**

**2. APPLICANT(S)**

| <b>Name</b>    | <b>Nationality</b> | <b>Address</b>                                          |
|----------------|--------------------|---------------------------------------------------------|
| SRM UNIVERSITY | INDIAN             | Amaravati, Mangalagiri,<br>Andhra Pradesh-522502, India |

**3. PREAMBLE TO THE DESCRIPTION**

The following specification particularly describes the invention and the manner in which it is to be performed.

## **FIELD**

The present disclosure relates, in general, to the field of battery charging management system. More specifically, embodiments of the present invention relate to a system and a method for managing charging infrastructure of electric vehicles.

## **BACKGROUND**

The background information herein below relates to the present disclosure but is not necessarily prior art.

The adoption of electric vehicles (EVs) has witnessed exponential growth in recent years, driven by advancements in technology, environmental awareness, and supportive governmental policies. However, the increasing number of EVs on the road places significant demands on the existing charging infrastructure, which predominantly relies on conventional grid-connected systems. These systems face critical challenges, including sudden load disturbances, limited access to real-time power demand data, high installation and operational costs, and inefficient load management.

Efforts to address these challenges have typically focused on incremental improvements to individual components of the charging infrastructure. While these approaches have contributed to marginal enhancements, they fail to offer a holistic solution that addresses the root causes of inefficiency and instability. Furthermore, the integration of renewable energy sources into EV charging operations, a promising avenue for reducing dependency on grid power, has been constrained by technical and operational limitations.

Thus, to tackle these issues, there was felt a need to provide a system that aims to enhance the efficiency and stability of EV charging infrastructure.

## **OBJECTS**

Some of the objects of the present disclosure, which at least one embodiment herein satisfies, are as follows.

It is an object of the present disclosure to ameliorate one or more problems of the prior art or to at least provide a useful alternative.

The main object of the present disclosure is to provide a system that is designed to operate under various conditions based on the availability of energy sources.

5       An object of the present disclosure is also to provide an invention for real-time control of large-scale power systems with the help of an online sequential learning algorithm-based neural network system for the purpose of commercialization in energy companies.

10       Another object of the present disclosure is to provide a system that is designed to automatically adjust its operational parameters based on real-time data inputs regarding energy availability and demand with the point of common coupling for the integration and automation.

15       Yet another object of the present disclosure is to provide a system that addresses critical issues such as grid disturbances and high operational costs by enabling adaptive responses to varying energy availability.

Still another object of the present disclosure is to provide a system that improves upon existing EV charging processes but also introduces a comprehensive solution that leverages IoT technology to create a more sustainable and economically viable charging infrastructure.

20       Other objects and advantages of the present disclosure will be more apparent from the following description when read in conjunction with the accompanying figures, which are not intended to limit the scope of the present disclosure.

## **SUMMARY**

25       The present disclosure relates, in general, to the field of electrical power systems. More specifically, embodiments of the present invention relate to a system and method for managing the charging infrastructure of electric vehicles.

The system broadly comprises a device layer, a network layer, an edge computing layer, a cloud layer, and an application layer. The system dynamically integrates

renewable energy sources, ESS, and grid power at a point of common coupling (PCC), enabling seamless switching between energy sources, optimized load distribution, reduced operational costs, and enhanced efficiency of EV charging operations.

5 The device layer is configured with a plurality of sensors for collecting real time data including energy availability from renewable energy sources, grid power, and energy storage systems (ESS), battery charging status including state of charge (SOC) and temperature, and load conditions of connected charging units, and a plurality of actuators for executing operational commands, wherein the actuators  
10 control the activation and deactivation of specific energy sources, and charging and discharging cycles of ESS.

The network layer is configured to facilitate communication between the device layer and subsequent layers by means of communication protocols selected from Message Queuing Telemetry Transport (MQTT) and Hypertext Transfer Protocol  
15 (HTTP), or a combination thereof, and further configured to transmit real-time data collected by the device layer and receive the control commands for the actuator operations.

The edge computing layer is configured to locally process the real-time data received from the network layer to determine operational parameters, wherein the  
20 processing includes analysing energy availability and load conditions, generating commands for optimal energy source switching based on predefined automation rules, including prioritizing solar power usage when available, activating grid power when ESS capacity falls below a threshold (20%); and balancing loads during high-demand periods; and transmitting processed data and decision outputs  
25 for advanced analytics and storage.

To optimize the charging conditions, the point of common coupling (PCC) is designed with the following conditions:

1. Solar – ON → ESS Charge → Grid OFF → Load ON
2. Solar – ON → ESS >80% → [Grid DC- AC] ON → Load ON

3. Solar – OFF → ESS ON → Grid OFF → Load ON
4. Solar – OFF → ESS <20% → [Grid DC- AC] ON → Load ON
5. Solar – OFF → ESS charge → [Grid AC- DC] ON → Load OFF

5 The cloud layer is configured to store the processed data received from the edge computing layer, further configured to analyze historical and real-time data using machine learning techniques to predict energy demand, optimize charging schedules, and identify maintenance requirements, and further configured to generate and send alerts for user action.

10 The application layer is configured to provide a user interface for monitoring energy usage, scheduling charging sessions, and customizing energy source preferences, and further configured to display actionable insights derived from the cloud layer, including maintenance notifications and cost-saving recommendations.

15 In an embodiment, the plurality of sensors in the device layer further comprises solar irradiance sensors to measure the energy output of solar panels, voltage and current sensors to monitor energy flow across the PCC, and thermal sensors to detect overheating in batteries or charging units.

20 In an embodiment, the actuators in the device layer include relays or switches to regulate the flow of energy between the grid, ESS, and charging stations and controllers for managing ESS charging and discharging cycles based on real-time load conditions.

In an embodiment, the network layer supports low-latency communication to facilitate rapid response to load changes at charging units and synchronization between the edge computing layer and the cloud layer.

25 In an embodiment, the edge computing layer employs embedded processors configured to execute real-time decision-making techniques, including rule-based logic and priority allocation techniques for energy management and pre-process data for transmission to the cloud layer to minimize bandwidth usage.

In an embodiment, the cloud layer implements machine learning techniques to predict energy availability from renewable sources based on weather forecasts and historical solar production data; and identify potential failures in charging stations by analyzing system health metrics collected by the sensors.

5 In an embodiment, the application layer further enables users to set charging priorities based on cost, energy source preference, or charging speed and receive recommendations for scheduling charging sessions during periods of high solar energy availability.

10 The present disclosure further envisages a method for managing energy distribution in an electric vehicle battery charging system. The method comprises the steps of :

- collecting real-time data in a device layer from a plurality of sensors, including energy availability, battery SOC, and load conditions,
- transmitting the collected data to an edge computing layer via a network layer,
- 15 • processing the data at the edge computing layer to determine energy source switching operations, and balance loads across charging units based on pre-defined automation rules,
- transmitting the processed data and operational commands to the device layer for execution and to the cloud layer for storage and analytics,
- 20 • analyzing, the data in the cloud layer to predict energy demand and availability, and optimize charging schedules based on user preferences, and
- displaying, real-time system status, analytics, and recommendations on a user interface in the application layer.

25 In an embodiment, the method further comprises dynamically switching energy sources based on pre-defined automation conditions, including enabling solar power as a primary source when available, activating grid power when ESS capacity is below 20%, and charging the ESS during periods of excess solar energy availability.

In an embodiment, the method further comprises generating predictive maintenance alerts in the cloud layer based on system performance metrics and transmitting the alerts to the application layer.

## **BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS**

5 The system and method for managing electric vehicle charging infrastructure will now be described with the help of the accompanying drawings, in which :

**Figure 1** illustrates a high-level block diagram of a system for managing electric vehicle charging infrastructure, in accordance with an embodiment of the present disclosure;

10 **Figure 2** illustrates a block diagram depicting the simulation of the battery charging management system on MATLAB, in accordance with an embodiment of the present disclosure;

**Figures 3A, 3B, and 3C** illustrate a block diagram depicting the central monitoring station responsible for controlling all charging units, in accordance with an embodiment of the present disclosure;

**Figure 4** illustrates a user interface created using Node-RED for data visualization, in accordance with an embodiment of the present disclosure;

**Figure 5** illustrates a dashboard that showcases how data is stored and processed in the cloud, in accordance with an embodiment of the present disclosure; and

20 **Figure 6** illustrates an Internet of Things (IoT) based robust bottle crushing method, in accordance with an embodiment of the present disclosure.

## **LIST OF REFERENCE NUMERALS USED IN THE DESCRIPTION AND DRAWINGS**

|      |                      |
|------|----------------------|
| 100  | SYSTEM               |
| 102  | DEVICE LAYER         |
| 102A | PLURALITY OF SENSORS |

|       |                             |
|-------|-----------------------------|
| 102AA | SOLAR IRRADIANCE SENSORS    |
| 102AB | VOLTAGE AND CURRENT SENSORS |
| 102AC | THERMAL SENSORS             |
| 102B  | PLURALITY OF ACTUATORS      |
| 104   | NETWORK LAYER               |
| 106   | EDGE COMPUTING LAYER        |
| 108   | CLOUD LAYER                 |
| 110   | APPLICATION LAYER           |

## DETAILED DESCRIPTION

Embodiments, of the present disclosure, will now be described with reference to the accompanying drawing.

5 Embodiments are provided so as to thoroughly and fully convey the scope of the present disclosure to the person skilled in the art. Numerous details are set forth, relating to specific components and methods to provide a complete understanding of the embodiments of the present disclosure. It will be apparent to the person skilled in the art that the details provided in the embodiments should not be  
10 construed to limit the scope of the present disclosure. In some embodiments, well-known apparatus structures, and well-known techniques are not described in detail.

The terminology used, in the present disclosure, is only for the purpose of explaining a particular embodiment and such terminology shall not be considered  
15 to limit the scope of the present disclosure. As used in the present disclosure, the forms “a”, “an”, and “the” may be intended to include the plural forms as well, unless the context clearly suggests otherwise. The terms, “comprises”, “comprising”, “including” and “having” are open-ended transitional phrases and therefore, specify the presence of stated features, integers, steps, operations,

elements, and/or components, but do not forbid the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

When an element is referred to as being, “embodied thereon”, “engaged to”, “coupled to” or “communicatively coupled to” another element, it may be directly on, engaged, connected, or coupled to the other element. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed elements.

The global adoption of electric vehicles (EVs) has experienced unprecedented growth over the past decade, spurred by technological advancements, increasing environmental awareness, and favorable governmental policies promoting the transition to sustainable transportation. This rapid proliferation of EVs has introduced new challenges to existing charging infrastructure, which was originally designed to support a limited number of vehicles and is largely dependent on conventional grid-connected systems.

Traditional charging infrastructure relies heavily on centralized power grids to meet the energy demands of EVs. This approach presents a series of critical challenges. Firstly, the sudden and unpredictable surges in power demand caused by the simultaneous charging of multiple EVs often result in load disturbances, straining the grid and leading to inefficiencies. Secondly, the lack of real-time data on power usage and demand limits the ability to optimize charging schedules and balance loads effectively, exacerbating grid instability. Additionally, the high costs associated with installing and maintaining conventional charging systems, particularly in remote or underserved areas, act as a significant barrier to widespread infrastructure deployment.

Efforts to address these limitations have primarily focused on incremental improvements to individual components of the charging ecosystem, such as the development of faster charging stations, enhanced energy storage solutions, and intelligent load management systems. While these initiatives have yielded some

benefits, they fail to provide an integrated solution that holistically addresses the inefficiencies and instabilities inherent in the current model of EV charging.

Furthermore, the integration of renewable energy sources, such as solar and wind, into EV charging operations has been recognized as a promising solution to reduce dependency on traditional grid power and mitigate environmental impact. However, this integration faces significant hurdles, including the intermittent nature of renewable energy generation, the lack of sophisticated energy management systems capable of balancing renewable input with grid supply, and the need for advanced storage solutions to ensure a reliable power supply during peak charging periods.

In light of these challenges, there is a pressing need for a novel system and method for managing EV charging infrastructure. Such a solution should aim to enhance the overall efficiency, reliability, and sustainability of charging operations by addressing both the technical and operational shortcomings of existing systems. Ideally, this system would seamlessly integrate renewable energy sources, leverage real-time data for intelligent load balancing, and reduce the reliance on centralized grid systems, thereby creating a more robust and future-ready charging infrastructure.

The present invention addresses these needs by providing an innovative approach to managing EV charging infrastructure, offering a comprehensive solution that improves system stability, optimizes energy utilization, and supports the widespread adoption of electric vehicles.

The present disclosure relates, in general, to the field of electrical power systems. More specifically, embodiments of the present invention relate to a system and method for managing charging infrastructure of electric vehicles.

The system proposed herein provides a significant improvement to existing charging systems for electric vehicles (EVs) by integrating an innovative IoT architecture that enhances efficiency and stability. While traditional EV charging solutions often rely on basic grid connections and lack real-time data

management, this disclosure introduces an optimal system that combines renewable energy sources, energy storage systems, and grid power to optimize charging operations.

This system disclosed herein is not merely a new application of existing technologies. It represents a novel approach to managing the complexities of EV charging. By employing a 5-layer IoT Architecture comprising device, network, edge computing, cloud, and application layers, i.e. the system allows for real-time monitoring, data analysis, and automation of charging processes. This results in reduced operational costs, improved load management, and minimized waste associated with energy consumption.

This invention introduces a groundbreaking smart energy distribution mechanism that dynamically allocates energy from multiple sources based on real-time demand and availability. This intelligent system ensures optimal resource utilization, reduces dependence on any single energy source, and enhances the resilience of the charging infrastructure. To further bolster efficiency, the invention integrates predictive maintenance algorithms that continuously monitor the health of charging stations, enabling proactive identification of potential failures. This minimizes downtime, reduces maintenance costs, and ensures consistent operational efficiency.

The system also allows user-driven customization, empowering users to set specific charging schedules and prioritize energy sources according to their preferences, thereby improving the user experience while optimizing energy usage. Seamless integration with smart grid technologies enables participation in demand response programs, allowing the system to adjust operations based on grid conditions, such as peak demand periods or excess renewable energy availability. This not only stabilizes the grid but also offers potential cost savings to users.

To tackle these issues, the system proposed herein aims to enhance the efficiency and stability of EV charging infrastructure. It integrates various elements, including hardware, software, and processes, to create a comprehensive solution

for managing the charging of EVs. The system combines renewable energy sources such as solar power and energy storage systems with grid power to optimize charging operations.

**Figure 1** illustrates a high-level block diagram of a system 100 for managing electric vehicle charging infrastructure, in accordance with an embodiment of the present disclosure.

The system 100 broadly comprises a device layer 102, a network layer 104, an edge computing layer 106, a cloud layer 108, and an application layer 110. The system dynamically integrates renewable energy sources, ESS, and grid power at a point of common coupling (PCC), enabling seamless switching between energy sources, optimized load distribution, reduced operational costs, and enhanced efficiency of EV charging operations.

The device layer 102 is configured with a plurality of sensors 102A for collecting real time data including energy availability from renewable energy sources, grid power, and energy storage systems (ESS), battery charging status including state of charge (SOC) and temperature, and load conditions of connected charging units, and a plurality of actuators 102B for executing operational commands, wherein the actuators control the activation and deactivation of specific energy sources, and charging and discharging cycles of ESS.

The network layer 104 is configured to facilitate communication between the device layer 102 and subsequent layers by means of communication protocols selected from Message Queuing Telemetry Transport (MQTT) and Hypertext Transfer Protocol (HTTP), or a combination thereof, and further configured to transmit real-time data collected by the device layer 102 and receive the control commands for the actuator operations.

The edge computing layer 106 is configured to locally process the real-time data received from the network layer 102 to determine operational parameters, wherein the processing includes analysing energy availability and load conditions, generating commands for optimal energy source switching based on predefined

automation rules, including prioritizing solar power usage when available, activating grid power when ESS capacity falls below a threshold (20%); and balancing loads during high-demand periods; and transmitting processed data and decision outputs for advanced analytics and storage.

5       The cloud layer 108 is configured to store the processed data received from the edge computing layer 106, further configured to analyze historical and real-time data using machine learning techniques to predict energy demand, optimize charging schedules, and identify maintenance requirements, and further configured to generate and send alerts for user action.

10       The application layer 110 is configured to provide a user interface for monitoring energy usage, scheduling charging sessions, and customizing energy source preferences, and is further configured to display actionable insights derived from the cloud layer 104, including maintenance notifications and cost-saving recommendations.

15       In an exemplary embodiment, the plurality of sensors 102A in the device layer 102 further comprises solar irradiance sensors 102AA to measure the energy output of solar panels, voltage and current sensors 102AB to monitor energy flow across the PCC, and thermal sensors 102AC to detect overheating in batteries or charging units.

20       In an exemplary embodiment, the actuators 102B in the device layer 102 include relays or switches 102BA to regulate the flow of energy between the grid, ESS, and charging stations and controllers 102BB for managing ESS charging and discharging cycles based on real-time load conditions.

25       In an exemplary embodiment, the network layer 102 supports low-latency communication to facilitate rapid response to load changes at charging units and synchronization between the edge computing layer 106 and the cloud layer 108.

In an exemplary embodiment, the edge computing layer 106 employs embedded processors configured to execute real-time decision-making techniques, including

rule-based logic and priority allocation techniques for energy management and pre-processing data for transmission to the cloud layer 108 to minimize bandwidth usage.

In an exemplary embodiment, the cloud layer 108 implements machine learning techniques to predict energy availability from renewable sources based on weather forecasts and historical solar production data; and identify potential failures in charging stations by analyzing system health metrics collected by the sensors.

In an exemplary embodiment, the application layer 110 further enables users to set charging priorities based on cost, energy source preference, or charging speed and receive recommendations for scheduling charging sessions during periods of high solar energy availability.

**Figure 2** illustrates a block diagram depicting the simulation of the battery charging management system on MATLAB, in accordance with an embodiment of the present disclosure. The integration of solar energy, energy storage systems (ESS), and grid connections at the Point of Common Coupling (PCC) highlights how these energy sources work together to automate charging based on availability and demand, supporting two main loads, i.e. a battery swapping station and four units of DC fast charging stations.

The system 100 is designed to seamlessly integrate renewable energy sources, conventional grid power, and energy storage solutions for efficient power management and distribution. It incorporates solar energy generation modules connected to a DC/DC converter to regulate output voltage, ensuring compatibility with downstream components. Additionally, a connection to the conventional electrical grid is established through a DC/AC converter, which stabilizes and adapts the power supply for the system's needs.

A Point of Common Coupling (PCC) serves as the central hub, enabling the integration of power from solar modules, the grid, and an energy storage system (ESS). This power is then distributed efficiently to various subsystems via a dedicated distribution unit. The ESS, comprising batteries and control circuitry,

stores surplus energy generated from the solar modules or grid. This stored energy is utilized during periods of peak demand or when the renewable energy supply is insufficient.

The system also features charging and swapping infrastructure to support electric vehicles (EVs). Fast charging stations draw power directly from the PCC, utilizing converters to adjust voltage and current according to EV requirements. Meanwhile, battery swapping stations accommodate multiple pre-charged battery modules, integrating with the central system to provide real-time status updates and optimize energy allocation.

**Figures 3A, 3B, and 3C** illustrate a block diagram depicting the central monitoring station in detail, which is responsible for controlling all charging units, in accordance with an embodiment of the present disclosure.

The Central Monitoring System (CMS) oversees the entire operation, offering a graphical user interface for real-time monitoring and control of energy flow, charging progress, and system health. Additionally, individual vehicle monitoring modules display specific vehicle-related information, ensuring comprehensive management and user accessibility.

The central monitoring and the user interface system for overseeing the generation, storage, and distribution of energy, as well as the parameters related to EV charging and battery health, ensure seamless integration and real-time status updates for efficient operation and maintenance of an electric vehicle charging infrastructure. The user interface allows users to monitor real-time data, track energy consumption, and manage charging schedules effectively, enhancing user engagement and operational efficiency and the cloud integration aspect of the system.

The energy management system 100 is supervised through a central monitoring station equipped with a user-friendly graphical interface, offering a comprehensive view of system operations. The interface includes features such as power generation control, which displays energy contributions from renewable

sources, e.g., solar, grid power, and energy storage systems (ESS). Key parameters like voltage, current, and power output are presented both, numerically and graphically. Real-Time status indicators monitor the performance and availability of solar panels, ESS, and grid inputs while displaying energy flow pathways and operational statuses. A control dashboard allows operators to manage energy allocation to sub-systems, such as charging stations or swapping modules.

For electric vehicles (EVs), the interface provides a detailed EV charging process overview, including information on required energy (in kWh), current and voltage levels during charging, estimated charging times, and power availability at charging stations. The system also visually represents EV connectivity to the infrastructure, highlighting data flow pathways.

In addition, the interface monitors critical EV battery charging parameters via a dedicated battery management system. It tracks voltage levels (alerting under-voltage conditions), detects over-current scenarios, monitors temperature to ensure safe operation, and provides fault detection with visual alerts, such as green and red signals, to notify operators of any anomalies.

**Figure 4** showcases how data is stored and processed in the cloud, in accordance with an embodiment of the present disclosure. It enables remote access to information, facilitates analytics, and supports decision-making processes based on real-time data from the charging infrastructure.

**Figure 5** illustrates a solar plant monitoring dashboard that showcases how data is stored and processed in the cloud, in accordance with an embodiment of the present disclosure.

The figure illustrates a design meant to provide real-time insights into various parameters related to solar power generation and output. The interface is divided into distinct sections, with each focusing on specific aspects of solar performance and metrics.

At the top of the interface, the title "Solar Plant Monitoring" is prominently displayed, emphasizing the system's primary purpose. The left-hand side panel serves as the navigation menu, allowing users to monitor various sub-systems within the setup. These options include 'Solar' which represents the solar power monitoring currently displayed; "EMS" (Energy Management System), which likely manages the overall energy flow; "PCC" (Point of Common Coupling), focusing on grid-related operations; "Battery Swapping," which tracks battery-based energy storage or replacement stations; and "Fast Charging," dedicated to monitoring infrastructure for fast-charging electric vehicles.

The central display panel serves as the dashboard, showcasing live data through two primary sections: 'Solar Power Generation' and 'Solar Output'. The Solar Power Generation section provides real-time data on critical parameters, including Solar\_Current (213.49 Volts), which indicates the voltage produced by the solar panels; Solar\_Temperature (26.05°C), a key metric as temperature affects panel efficiency; Solar\_Power (123.61 Watts), representing the total power generated; and Solar\_Current (123.61 Amps), reflecting the current flow in the system.

The Solar Output section focuses on the power delivered after initial processing. It highlights metrics such as Solar\_out\_Voltage (432 Volts), which is the output voltage supplied to the grid or connected systems; Solar\_Irradiation, showing the intensity of sunlight, crucial for efficiency analysis; Solar\_out\_Power (40 Watts), representing the power output after internal conversions or losses; and Solar\_out\_Current (94 Amps), indicating the current delivered as part of the final output. Together, these components provide a detailed and organized view of the solar plant's performance.

**Figure 6** illustrates an Internet of Things (IoT) based robust bottle crushing method 600, in accordance with an embodiment of the present disclosure. The order in which the method 600 is described is not intended to be construed as a limitation, and any number of the described method blocks may be combined in

any order to implement the method 600, or an alternative method. The method 600 includes the following steps:

5 In method step 602, the method comprises collecting real-time data in a device layer from a plurality of sensors, including energy availability, battery SOC, and load conditions,

In method step 604, the method comprises transmitting the collected data to an edge computing layer via a network layer,

10 In method step 606, the method comprises processing the data at the edge computing layer to determine energy source switching operations, and balance loads across charging units based on pre-defined automation rules,

In method step 608, the method comprises transmitting the processed data and operational commands to the device layer for execution and to the cloud layer for storage and analytics,

15 In method step 610, the method comprises analyzing, the data in the cloud layer to predict energy demand and availability and optimize charging schedules based on user preferences, and

In method step 612, the method comprises displaying, real-time system status, analytics, and recommendations on a user interface in the application layer.

20 In an exemplary embodiment, the method further comprises dynamically switching energy sources based on pre-defined automation conditions, including enabling solar power as a primary source when available, activating grid power when ESS capacity is below 20%, and charging the ESS during periods of excess solar energy availability.

25 In an exemplary embodiment, the method further comprises generating predictive maintenance alerts in the cloud layer based on system performance metrics and transmitting the alerts to the application layer.

**Experimental Analysis:**

The experimental simulations collectively demonstrate the functionality and effectiveness of the proposed IoT architecture in managing battery charging systems while optimizing energy utilization from multiple sources. The results reveal significant improvements in efficiency, cost savings, and user experience within the electric vehicle charging infrastructure.

The experiments initially focused on integrating diverse energy sources, including solar energy, energy storage systems (ESS), and grid power. The solar panels exhibited a daily energy output of 1032 kW (56.8%), while the ESS contributed 576 kW (31.7%), and the grid supplied 209 kW (11.5%), providing critical insights into the overall energy generation capabilities of the system.

To manage energy distribution effectively, four automation conditions were established at the point of common coupling (PCC), allowing the system to adapt dynamically to varying energy availability and demand. The testing included two primary loads: a battery swapping station and four DC fast charging stations. The battery swapping station was designed to charge 90 batteries daily, consuming 54 KW of energy, while the DC fast charging stations were assessed under various vehicle charging parameters.

A central monitoring station utilizing Node-RED for data visualization was implemented, enabling real-time monitoring of energy consumption and system performance.

Additionally, cloud integration with AWS was tested to ensure secure storage and remote accessibility of data, thereby facilitating effective decision-making based on real-time analytics.

The efficiency of renewable energy usage was evaluated, showing an energy-saving efficiency of 88.6%. Cost-saving efficiency, derived from operational cost analysis, was calculated to be 88.5%.

Furthermore, the revenue potential of the battery swapping station and DC fast charging stations was analyzed, estimating a daily revenue generation of approximately ₹29,494 after accounting for grid power costs.

## TECHNICAL ADVANCEMENTS AND ECONOMIC SIGNIFICANCE

The present disclosure described herein above has several technical advantages including, but not limited to, the realization of a system and method for managing charging infrastructure of electric vehicles, which:

- 5       • improves the efficiency of the entire charging process of an electric vehicle thereby reducing wait times and energy waste,
- lowers the operational costs associated with charging an electric vehicle utilizing renewable energy sources,
- contributes to a more sustainable transportation ecosystem, thereby reducing  
10       greenhouse gas emissions and promoting cleaner air quality,
- enhances the experience of users by providing flexibility in how and when they charge their vehicles,
- stabilizes the electrical grid during peak demand periods, thereby contributing to overall grid reliability and efficiency,
- 15       • addresses critical issues such as grid disturbances and high operational costs by enabling adaptive responses to varying energy availability,
- allows for easy scalability as the number of electric vehicles and charging stations increases, making it suitable for both small-scale deployments and large-scale infrastructure projects.

20       The embodiments herein and the various features and advantageous details thereof are explained with reference to the non-limiting embodiments in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways  
25       in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

The foregoing description of the specific embodiments so fully reveals the general nature of the embodiments herein that others can, by applying current knowledge,  
30       readily modify and/or adapt for various applications such specific embodiments

without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope of the embodiments as described herein.

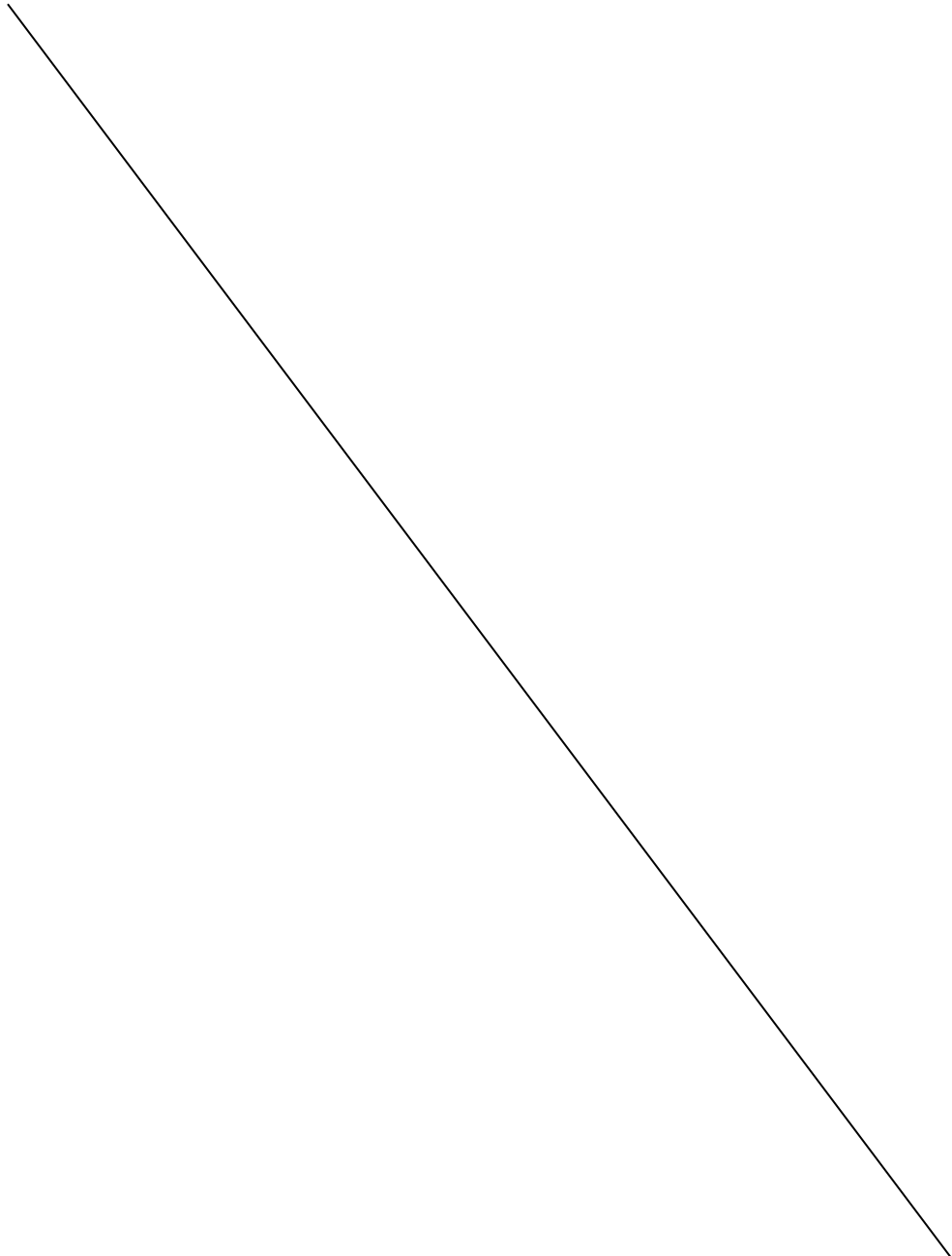
Throughout this specification, the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, or group of elements, but not the exclusion of any other element, or group of elements.

Any discussion of documents, acts, materials, devices, articles or the like that has been included in this specification is solely for the purpose of providing a context for the disclosure. It is not to be taken as an admission that any or all of these matters form a part of the prior art base or were common general knowledge in the field relevant to the disclosure as it existed anywhere before the priority date of this application.

The numerical values mentioned for the various physical parameters, dimensions or quantities are only approximations and it is envisaged that the values higher/lower than the numerical values assigned to the parameters, dimensions or quantities fall within the scope of the disclosure, unless there is a statement in the specification specific to the contrary.

While considerable emphasis has been placed herein on the components and component parts of the preferred embodiments, it will be appreciated that many embodiments can be made and that many changes can be made in the preferred embodiments without departing from the principles of the disclosure. These and other changes in the preferred embodiment as well as other embodiments of the disclosure will be apparent to those skilled in the art from the disclosure herein,

whereby it is to be distinctly understood that the foregoing descriptive matter is to be interpreted merely as illustrative of the disclosure and not as a limitation.



**WE CLAIM :**

1. A system (100) for managing electric vehicle (EV) charging infrastructure, comprising:

a device layer (102) configured with :

5 a plurality of sensors (102A) for collecting real time data including

(i) energy availability from renewable energy sources, grid power, and energy storage systems (ESS),

10 (ii) battery charging status including state of charge (SOC) and temperature, and

(iii) load conditions of connected charging units, and

a plurality of actuators (102B) for executing operational commands, wherein the actuators control

15 (i) the activation and deactivation of specific energy sources, and

(ii) charging and discharging cycles of ESS;

20 a network layer (104) configured to facilitate communication between the device layer (102) and subsequent layers by means of communication protocols selected from Message Queuing Telemetry Transport (MQTT) and Hypertext Transfer Protocol (HTTP), or a combination thereof, and further configured to transmit real-time data collected by the device layer (102) and receive the control commands for the actuator operations;

an edge computing layer (106) configured to :

25 locally process the real-time data received from the network layer (102) to determine operational parameters, wherein the processing includes :

analysing energy availability and load conditions,

generating commands for optimal energy source switching based on predefined automation rules, including:

- prioritizing solar power usage when available;
- activating grid power when ESS capacity falls below a threshold (20%); and
- balancing loads during high-demand periods; and
- transmitting processed data and decision outputs for advanced analytics and storage;

a cloud layer (108) configured to store the processed data received from the edge computing layer (106), further configured to analyze historical and real-time data using machine learning techniques to predict energy demand, optimize charging schedules, and identify maintenance requirements, and further configured to generate and send alerts for user action; and

an application layer (110) configured to provide a user interface for monitoring energy usage, scheduling charging sessions, and customizing energy source preferences, and further configured to display actionable insights derived from the cloud layer (104), including maintenance notifications and cost-saving recommendations,

wherein the system (100) dynamically integrates renewable energy sources, ESS, and grid power at a point of common coupling (PCC), enabling seamless switching between energy sources, optimized load distribution, reduced operational costs, and enhanced efficiency of EV charging operations.

2. The system (100) as claimed in claim 1, wherein the plurality of sensors (102A) in the device layer (102) further comprises:

solar irradiance sensors (102AA) to measure the energy output of solar panels;

voltage and current sensors (102AB) to monitor energy flow across the PCC; and

thermal sensors (102AC) to detect overheating in batteries or charging units.

- 5           3. The system (100) as claimed in claim 1, wherein the actuators (102B) in the device layer (102) include:

relays or switches (102BA) to regulate the flow of energy between the grid, ESS, and charging stations; and

- 10                   controllers (102BB) for managing ESS charging and discharging cycles based on real-time load conditions.

4. The system (100) as claimed in claim 1, wherein the network layer (102) supports low-latency communication to facilitate rapid response to load changes at charging units and synchronization between the edge computing layer (106) and the cloud layer (108).

- 15           5. The system (100) as claimed in claim 1, wherein the edge computing layer (106) employs embedded processors configured to:

execute real-time decision-making algorithms, including rule-based logic and priority allocation algorithms for energy management; and

- 20                   pre-process data for transmission to the cloud layer (108) to minimize bandwidth usage.

6. The system (100) as claimed in claim 1, wherein the cloud layer (108) implements machine learning algorithms to:

predict energy availability from renewable sources based on weather forecasts and historical solar production data; and

- 25                   identify potential failures in charging stations by analyzing system health metrics collected by the sensors.

7. The system (100) as claimed in claim 1, wherein the application layer (110) further enables users to:

set charging priorities based on cost, energy source preference, or charging speed; and

5 receive recommendations for scheduling charging sessions during periods of high solar energy availability.

8. A method (600) for managing energy distribution in an electric vehicle battery charging system, comprising:

10 collecting (602), from a plurality of sensors (102A), real-time data in a device layer (102), including energy availability, battery SOC, and load conditions,

transmitting (604), via a network layer (104), the collected data to an edge computing layer (106);

15 processing (606), the data at the edge computing layer (106) to determine energy source switching operations, and balance loads across charging units based on pre-defined automation rules;

transmitting (608), the processed data and operational commands to the device layer (102) for execution and to the cloud layer (108) for storage and analytics;

20 analyzing (610), the data in the cloud layer (108) to predict energy demand and availability, and optimize charging schedules based on user preferences; and

displaying (612), real-time system status, analytics, and recommendations on a user interface in the application layer (110).

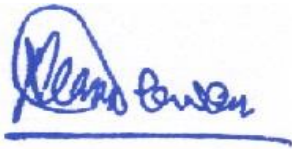
25 9. The method as claimed in claim 8, further comprises dynamically switching energy sources based on pre-defined automation conditions, including:

enabling solar power as a primary source when available;

activating grid power when ESS capacity is below 20%; and  
charging the ESS during periods of excess solar energy availability.

- 5 10. The method as claimed in claim 8 further comprises generating predictive maintenance alerts in the cloud layer (108) based on system performance metrics and transmitting the alerts to the application layer (110).

**Dated this 05<sup>th</sup> day of February, 2025**



---

**MOHAN RAJKUMAR DEWAN, IN/PA – 25  
OF R. K. DEWAN & CO.  
AUTHORIZED AGENT OF APPLICANT**

**TO,  
THE CONTROLLER OF PATENTS  
THE PATENT OFFICE, CHENNAI**

## ABSTRACT

### SYSTEM AND METHOD FOR MANAGING CHARGING INFRASTRUCTURE OF ELECTRIC VEHICLES THROUGH IoT

The present disclosure envisages a system (100) and method (1000) for managing electric vehicle (EV) charging infrastructure. The system broadly comprises a device layer (102), a network layer (104), an edge computing layer (106), a cloud layer (108), and an application layer (110). The device layer (102) is equipped with sensors to collect real-time data on renewable energy availability, grid power, energy storage systems (ESS), battery state of charge (SOC), temperature, and load conditions, as well as actuators to manage energy sources and ESS charging or discharging. The network layer (104) facilitates seamless data exchange through protocols like MQTT and HTTP, enabling the transmission of sensor data and reception of actuator commands. The edge computing layer (106) processes data locally to optimize energy source switching, prioritize solar power usage, activate grid power when ESS capacity is low, and balance loads during peak demand periods. The cloud layer (108) stores and analyzes data using machine learning to predict energy demand, optimize energy schedules, and identify maintenance requirements while generating alerts. The application layer (110) provides a user interface for monitoring energy usage, scheduling operations, customizing preferences, and delivering actionable insights.

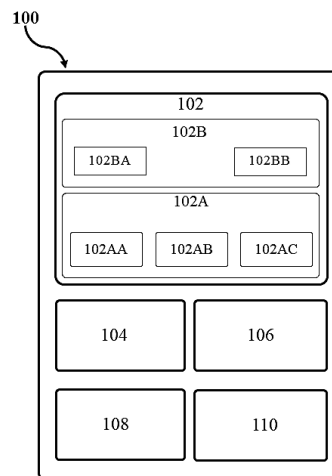


Figure 1